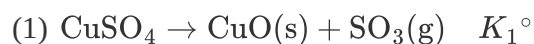


Question

It is known that anhydrous copper sulfate decomposes at high temperature according to the following two equations:



1. Now, fill a 1.00 L closed container with argon gas at 273 K and 100 mmHg column pressure, and add a weighed amount of anhydrous copper sulfate solid. Heat to 1050 K and when the reaction reaches equilibrium, the mass of the solid decreases by 0.3869 g, and the pressure in the closed container is 108.59kPa. Calculate K_1° and K_2° at this temperature.

2. Assuming that the difference in standard molar reaction enthalpy between reaction (1) and reaction (2) is $\Delta_r H_m^\circ(1) - \Delta_r H_m^\circ(2) = 27 \text{ kJ/mol}$, calculate the temperature at which the partial pressure of SO_2 is 5 times the partial pressure of SO_3 when equilibrium is reached.

3. At higher temperatures ($>1400 \text{ K}$), the mass of the solid will further decrease. At the same certain temperature ($>1400 \text{ K}$), due to the generation of new reactions, have the partial pressures of various gases increased or decreased compared to the original?

Answer the above questions and select the correct option.

A. All other options are incorrect

B.

1. $K_1^\circ = 0.120$,
 $K_2^\circ = 0.949$
2. $T = 6.3 \times 10^2 \text{ K}$
3. $p(\text{O}_2)$ increases, $p(\text{SO}_2)$ increases, $p(\text{SO}_3)$ decreases

C.

1. $K_1^\circ = 0.120$,
 $K_2^\circ = 0.949$

2. $T = 6.3 \times 10^2 \text{ K}$

3. $p(\text{O}_2)$ increases, $p(\text{SO}_2)$ decreases, $p(\text{SO}_3)$ decreases

D. 1. $K_1^\circ = 0.120$,
 $K_2^\circ = 0.949$

2. $T = 6.3 \times 10^2 \text{ K}$

3. $p(\text{O}_2)$ increases, $p(\text{SO}_2)$ decreases, $p(\text{SO}_3)$ remains unchanged

E. 1. $K_1^\circ = 0.120$,
 $K_2^\circ = 0.949$

2. $T = 3.3 \times 10^2 \text{ K}$

3. $p(\text{O}_2)$ increases, $p(\text{SO}_2)$ decreases, $p(\text{SO}_3)$ remains unchanged

F. 1. $K_1^\circ = 0.120$,
 $K_2^\circ = 0.949$

2. $T = 3.3 \times 10^2 \text{ K}$

3. $p(\text{O}_2)$ increases, $p(\text{SO}_2)$ increases, $p(\text{SO}_3)$ increases

G. 1. $K_1^\circ = 0.120$,
 $K_2^\circ = 0.949$

2. $T = 3.3 \times 10^2 \text{ K}$

3. $p(\text{O}_2)$ increases, $p(\text{SO}_2)$ decreases, $p(\text{SO}_3)$ increases

H. 1. $K_1^\circ = 12.0$,
 $K_2^\circ = 94.9$

2. $T = 6.3 \times 10^2 \text{ K}$

3. $p(\text{O}_2)$ increases, $p(\text{SO}_2)$ decreases, $p(\text{SO}_3)$ remains unchanged

I. 1. $K_1^\circ = 12.0$,

$$K_2^\circ = 94.9$$

$$2. T = 3.3 \times 10^2 \text{ K}$$

3. $p(\text{O}_2)$ increases, $p(\text{SO}_2)$ decreases, $p(\text{SO}_3)$ decreases

J. 1. $K_1^\circ = 12.0$,

$$K_2^\circ = 94.9$$

$$2. T = 3.3 \times 10^2 \text{ K}$$

3. $p(\text{O}_2)$ increases, $p(\text{SO}_2)$ decreases, $p(\text{SO}_3)$ increases

Answer

Correct Answer: **D**

Detailed Explanation

$$n(\text{Ar}) = p_0 V / RT_0 = 100 \times 1.00 \times 101.325 / (8.314 \times 760 \times 273) = 5.87 \times 10^{-3} \text{ (mol)}$$

$$\text{At } 1050 \text{ K, } p(\text{Ar}) = nRT/V = 5.87 \times 10^{-3} \times 1050 \times 8.314 / 1.00 = 51.3 \text{ (kPa)}$$

CHECKPOINT

1 PTS

$$p(\text{Ar}) = 51.3 \text{ kPa}$$

$$\text{Therefore: } p(\text{SO}_2) + p(\text{O}_2) + p(\text{SO}_3) = 108.59 - 51.3 = 57.3 \text{ (kPa)}$$

According to the conservation of mass,

$$m(\text{SO}_2) + m(\text{O}_2) + m(\text{SO}_3) = V [M(\text{SO}_2)p(\text{SO}_2) + M(\text{O}_2)p(\text{O}_2) + M(\text{SO}_3)p(\text{SO}_3)] / RT = 0.3869 \text{ g}$$

$$p(\text{SO}_2) = 2p(\text{O}_2)$$

CHECKPOINT

0.5 PTS

$$p(\text{SO}_2) = 2p(\text{O}_2)$$

Solving the three equations for three unknowns, we get:

$$p(\text{SO}_2) = 30.2\text{kPa}$$

$$p(\text{O}_2) = 15.1\text{kPa}$$

$$p(\text{SO}_3) = 12.0\text{kPa}$$

CHECKPOINT

1.5 PTS

$$p(\text{SO}_2) = 30.2\text{kPa}, p(\text{O}_2) = 15.1\text{kPa}, p(\text{SO}_3) = 12.0\text{kPa}$$

$$\text{Thus } K_1^\circ = p(\text{SO}_3)/p^\circ = 0.120$$

$$K_2^\circ = p^2(\text{SO}_2)p(\text{O}_2)/p^\circ p(\text{SO}_3) = 0.949$$

CHECKPOINT

0.5 PTS

$$K_1^\circ = 0.120, K_2^\circ = 0.949$$

The second question is based on the Van't Hoff equation:

$$\ln \frac{K_1^\circ(T)}{K_1^\circ(1050\text{ K})} = \frac{\Delta_r H_m^\circ(1)}{R} \left(\frac{1}{1050\text{ K}} - \frac{1}{T} \right)$$

$$\ln \frac{K_2^\circ(T)}{K_2^\circ(1050\text{ K})} = \frac{\Delta_r H_m^\circ(2)}{R} \left(\frac{1}{1050\text{ K}} - \frac{1}{T} \right)$$

Subtracting the two equations and rearranging slightly, we have:

$$\ln \frac{K_2^\circ(1050\text{ K})}{K_1^\circ(1050\text{ K})} - \ln \frac{K_2^\circ(T)}{K_1^\circ(T)} = \frac{\Delta_r H_m^\circ(1) - \Delta_r H_m^\circ(2)}{R} \left(\frac{1}{1050\text{ K}} - \frac{1}{T} \right)$$

According to the question, at this temperature:

$$p(\text{SO}_2) = 5p(\text{SO}_3)$$

In addition, under the initial conditions, $p(\text{SO}_2) = 2p(\text{O}_2)$. In reaction (2), the conversion rate of SO_2 and the conversion rate of O_2 are in a ratio of 2 : 1, which is the same as the initial ratio. Therefore, the relationship $p(\text{SO}_2) = 2p(\text{O}_2)$ always holds.

Then:

$$\frac{K_2^\circ(T)}{K_1^\circ(T)} = \frac{p_{\text{SO}_2}^2 p_{\text{O}_2}}{p_{\text{SO}_3}^2} = \frac{5^2 \times \frac{5}{2}}{1} = 62.5$$

Substituting all known data, we have

$$\ln \frac{0.949}{0.120} - \ln 62.5 = \frac{27 \times 10^3}{8.314} \left(\frac{1}{1050} - \frac{1}{T} \right)$$

Solving for T , we get $T = 6.3 \times 10^2 \text{ K}$

CHECKPOINT

2 PTS

$$T = 6.3 \times 10^2 \text{ K}$$

At higher temperatures ($>1400 \text{ K}$), the mass of the solid will further decrease, producing a new reaction. At this time, the only gas produced by the decomposition reaction can be oxygen. The chemical reaction equation for this step is:



The new reaction occurring at this time is equivalent to adding extra oxygen to the original equilibrium. According to Le Chatelier's principle, the equilibrium of reaction (2) in the question shifts in the reverse direction, so $p(\text{O}_2)$ increases and $p(\text{SO}_2)$ decreases (these two substances are only related to the equilibrium of reaction (2)).

However, since the temperature remains constant, the equilibrium constant of the equilibrium remains unchanged, and $p(\text{SO}_3) = K_1^\circ p^\circ$, so $p(\text{SO}_3)$ remains unchanged.

CHECKPOINT

0.5 PTS

 $p(\text{O}_2)$ increases

CHECKPOINT

0.5 PTS

 $p(\text{SO}_2)$ decreases

CHECKPOINT

0.5 PTS

 $p(\text{SO}_3)$ remains unchanged